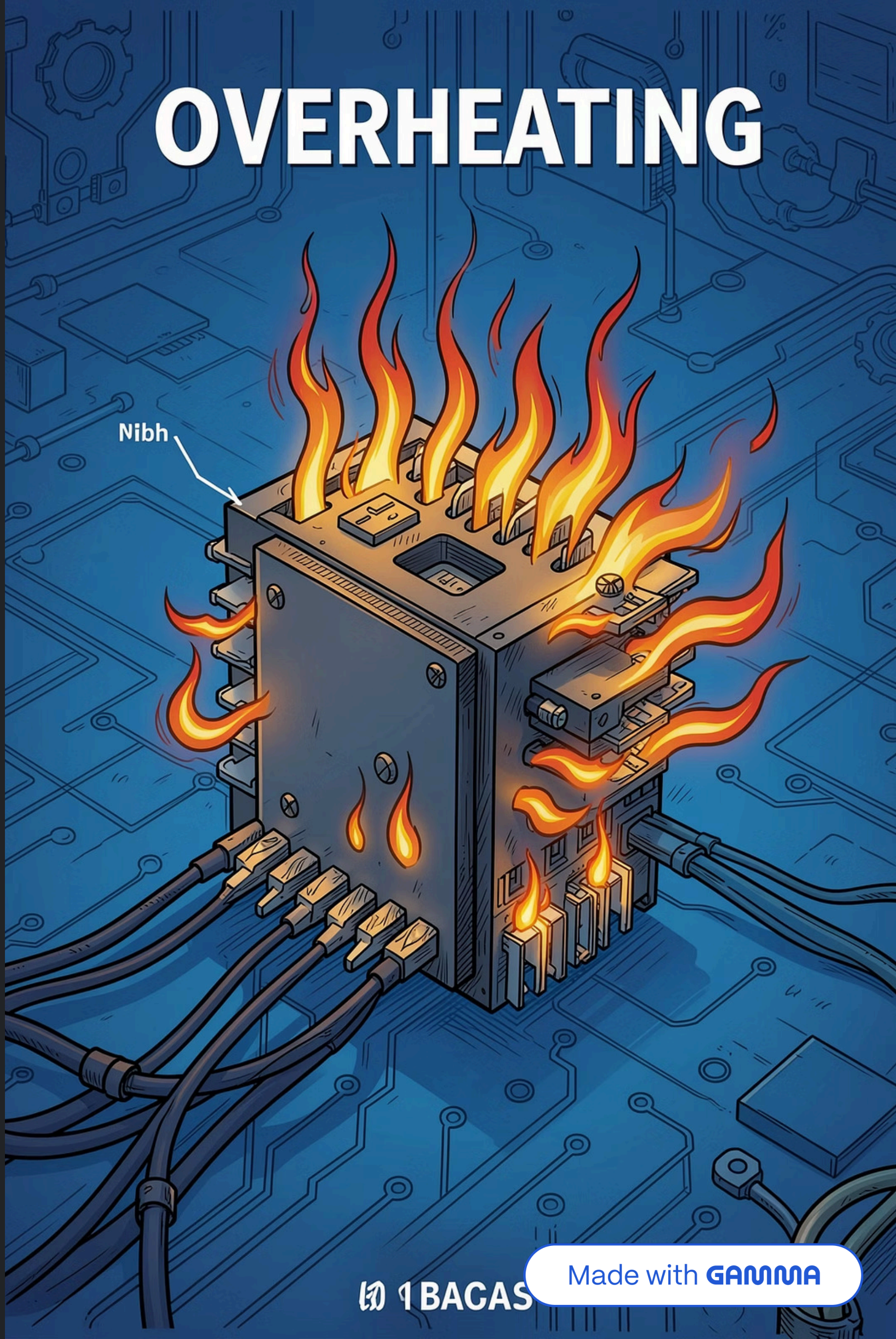
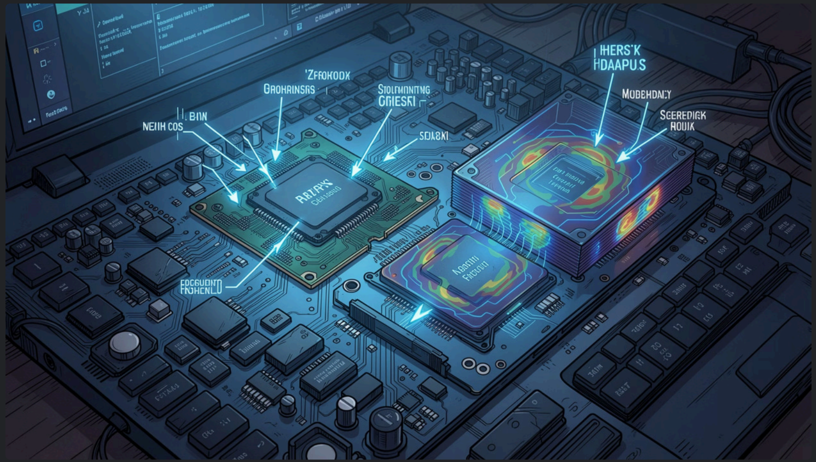
Overheating in Electronic Systems: Nazila Aliyeva Verified by: Physics Teacher Azerbaijan Telman Askeraliyev (Fizika muellimi, Azerbaijan, Baku)

What Is Overheating?

Overheating occurs when a device's internal temperature exceeds its safe operating range, causing performance degradation, accelerated aging, or immediate failure. For semiconductors this typically means junction temperatures above specified TJ(max).

- Short-term: thermal throttling, transient faults
- Long-term: accelerated wear-out, electromigration
- Safety: ignition risk in extreme cases





Primary Causes of Overheating

Heat is produced wherever electrical energy is dissipated. Common sources in systems:

- High current through resistive traces and shunts
- Power-dense ICs: CPUs, GPUs, FPGAs (dynamic and static power)
- Electromechanical: motors, relays, solenoids (copper losses + friction)
- Power semiconductors: MOSFETs, BJTs, diodes (switching & conduction losses)
- Poor thermal design: restricted airflow, inadequate heatsinking, high-impedance thermal paths

Heat Generation Mechanisms

At the component and PCB level, the main mechanisms are:

Conduction Losses

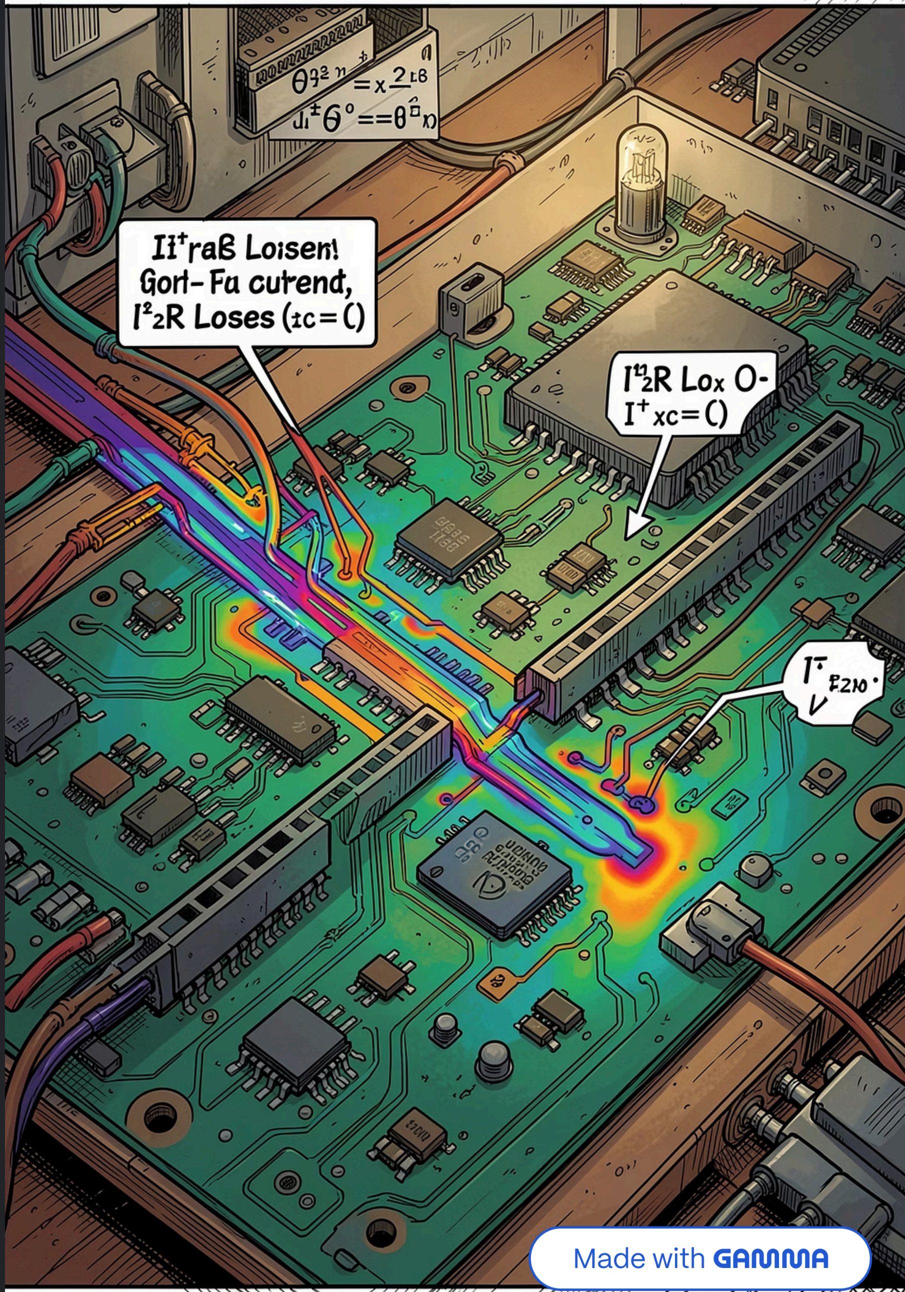
Resistive heating: P = I^2·R in traces, vias, windings. Localized hotspots form at high-current density areas.

Switching Losses

MOSFET/BJT transitions during switching produce energy dissipation proportional to voltage × current × switching time.

Leakage & Static Power

Subthreshold leakage and bias currents in modern ICs contribute continuous heat, especially at elevated ambient temperatures.



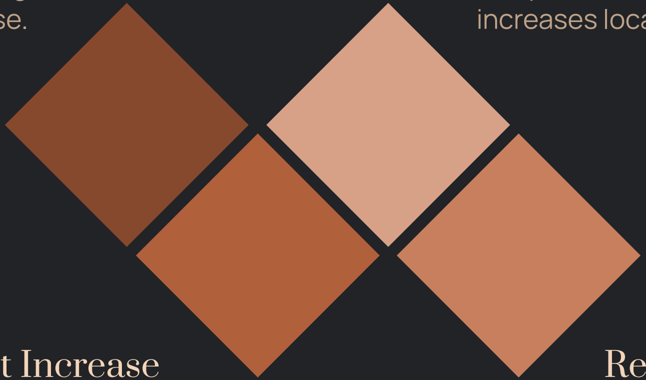
From Heat to Failure

Power Dissipation

I^2R losses and switching losses rise.

Local Heating

Temperature at hotspots increases locally.



Current Increase

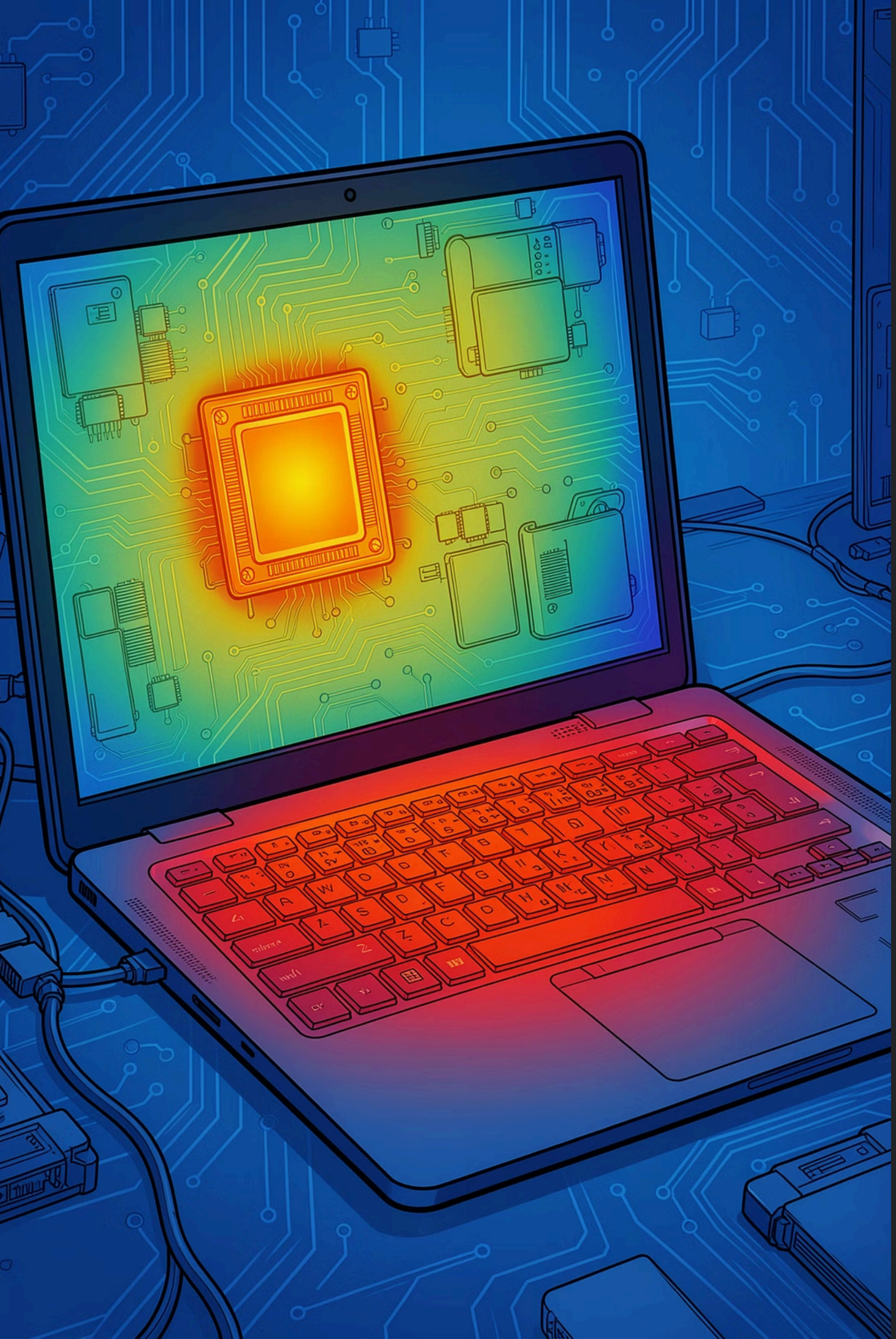
Higher electrical current through conductors.

Reliability Degradation

Elevated temp accelerates failure mechanisms.

Elevated temperatures accelerate multiple failure mechanisms:

- Electromigration in metal traces (high current density + temperature)
- Die attach degradation and solder joint fatigue
- Semiconductor parameter shifts: threshold voltage, mobility, leakage
- Capacitor electrolyte drying and reduced capacitance

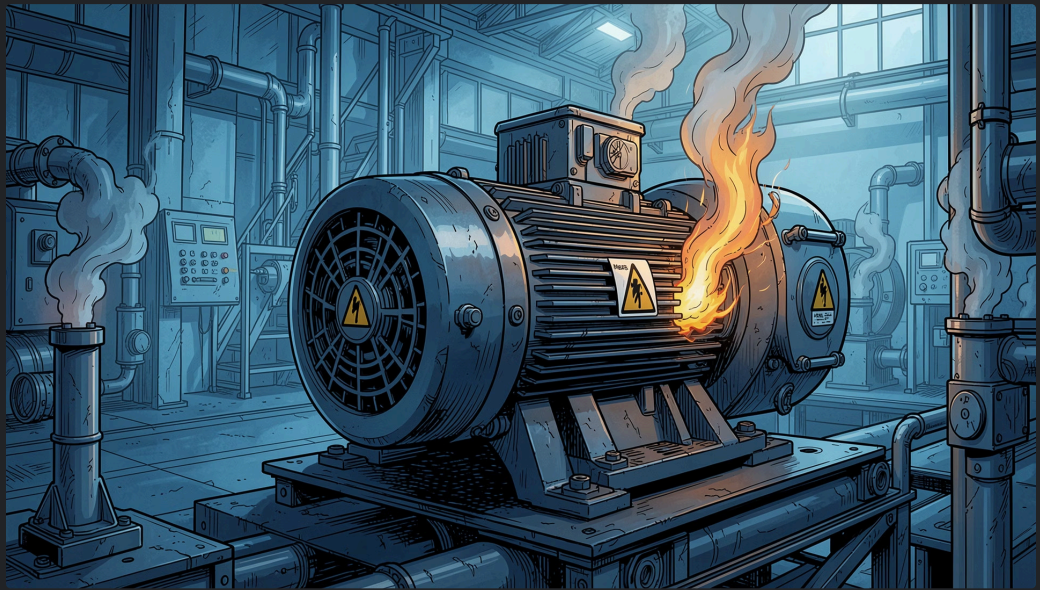


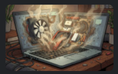
System-Level Impacts

Consequences at system level include:

- Performance loss: thermal throttling, timing violations
- Reduced MTBF: faster wear and earlier end-of-life
- Functional failures: brownouts, latch-up, memory errors
- Safety hazards: battery thermal runaway, enclosure surface burns

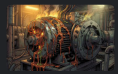
Real-World Examples





Laptops

Dust-clogged fans + aging thermal paste → CPU/GPU throttling, battery heating.



Motors

Overcurrent and stalled conditions cause winding insulation breakdown and mechanical damage.

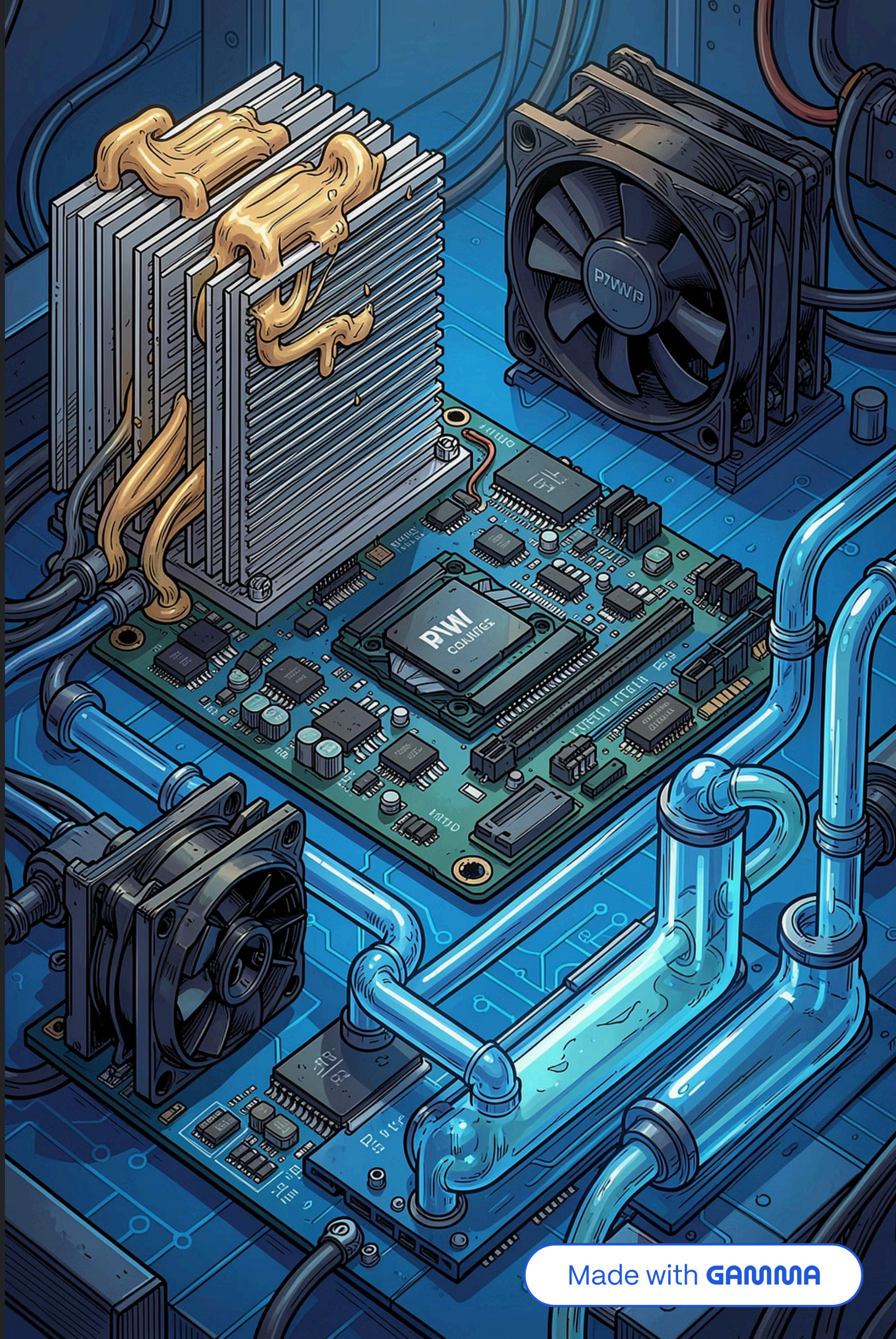
Industrial Power Electronics

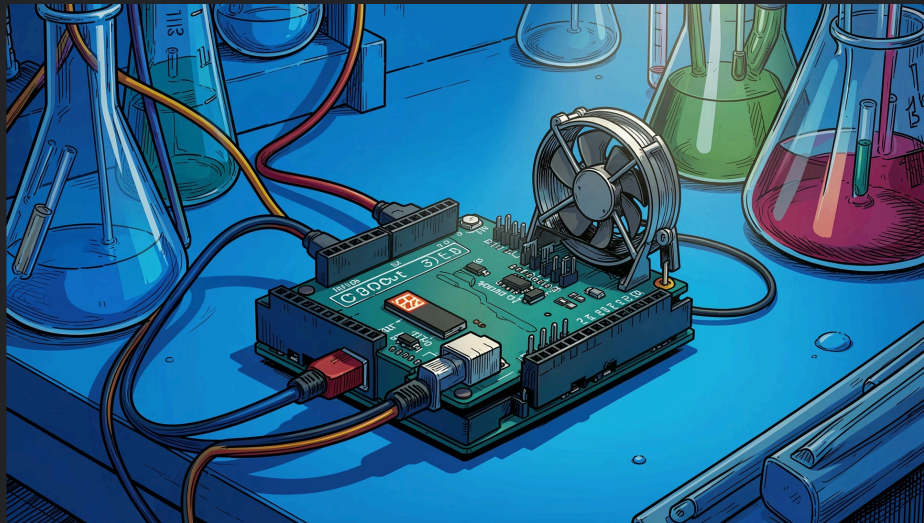
Poor cooling and high ambient temps lead to capacitor drying, PCB delamination, and relay welds.

Basic Cooling Methods

Engineering controls to remove or spread heat:

- Passive: heatsinks, thermal pads, copper pours, PCB thermal vias
- Active: fans (PWM speed control), blowers, forced convection
- Advanced: liquid cooling, vapor chambers, Peltier modules (thermoelectric)
- Thermal interface optimization: low-thermal-resistance TIMs, correct mounting pressure





Simple Protection Techniques (Circuit-Level)

Practical, low-cost methods to limit temperature-related risk:

Current Sensing & Limiting

Use accurate shunts or Hall sensors with current-limit loops (ADC + MCU or analog comparator) to prevent sustained I^2R heating.

Thermal Cutoffs & PTC/NTC

Employ thermal fuses, NTC inrush limiters, or PTC resettable fuses in series with high-power paths.

Temperature Sensing & Control

Place thermistors or digital temp sensors near hotspots and implement closed-loop fan/heatsink control (PID or PWM throttling).

Power Sequencing & Soft-Start

Smooth startup reduces surge heating: controlled slew rates, soft-start circuits, pre-charge resistors.

Design Checklist & Next Steps

Concise engineering actions to reduce overheating risk before production:

- Run thermal simulations and validate with IR imaging/thermocouples under worst-case loads
- Optimize PCB layout: copper pours, width of high-current traces, thermal vias under components
- Specify components with adequate TJ(max) and derating margins; verify SOA for power semiconductors
- Include monitoring and fail-safe logic: thermal shutdown, graceful degradation modes
- Plan maintenance: accessible fans, filter elements, and clear airflow paths